LINUX ADMINISTRATION LAB SHEET - 3

Q1. Hello Script

```
juhee@kali:~/linux_lab3$ cat hello.sh
#!/bin/bash
NAME="Juhee"
echo "Hello, $NAME! Welcome to Linux Shell Scripting Lab."

juhee@kali:~/linux_lab3$ chmod +x hello.sh
juhee@kali:~/linux_lab3$ ./hello.sh
Hello, Juhee! Welcome to Linux Shell Scripting Lab.
juhee@kali:~/linux_lab3$ bash hello.sh
Hello, Juhee! Welcome to Linux Shell Scripting Lab.
```

Q2. User Input Script

```
juhee@kali:~/linux_lab3$ cat user_input.sh
#!/bin/bash
read -p "Enter Student Name: " name
read -p "Enter Branch: " branch
read -p "Enter Semester: " sem
echo "----------------------------------------"
echo "Welcome $name! Branch: $branch | Semester: $sem"
echo "----------------------------------------"

juhee@kali:~/linux_lab3$ chmod +x user_input.sh
juhee@kali:~/linux_lab3$ ./user_input.sh
Enter Student Name: Juhee
Enter Branch: CSE
Enter Semester: 5th
----------------------------------------
Welcome Juhee! Branch: CSE | Semester: 5th
----------------------------------------
```

Q3. Arithmetic Calculator

```
juhee@kali:~/linux_lab3$ cat calculator.sh
#!/bin/bash
read -p "Enter first integer: " num1
read -p "Enter second integer: " num2

sum=$((num1 + num2))
diff=$((num1 - num2))
prod=$((num1 * num2))
quot=$((num1 / num2))

echo "Sum: $sum"
echo "Difference: $diff"
echo "Product: $prod"
echo "Quotient: $quot"

juhee@kali:~/linux_lab3$ chmod +x calculator.sh
juhee@kali:~/linux_lab3$ ./calculator.sh
Enter first integer: 20
Enter second integer: 4
Sum: 24
Difference: 16
Product: 80
Quotient: 5
```

Q4. Command Substitution

```
juhee@kali:~/linux_lab3$ cat system_info.sh
#!/bin/bash
CURRENT_DATE=$(date)
CURRENT_DIR=$(pwd)
LOGGED_USER=$(whoami)

echo "Current Date & Time: $CURRENT_DATE"
echo "Current Directory  : $CURRENT_DIR"
echo "Logged-in User     : $LOGGED_USER"

juhee@kali:~/linux_lab3$ chmod +x system_info.sh
juhee@kali:~/linux_lab3$ ./system_info.sh
Current Date & Time: Mon Nov  2 11:47:00 IST 2026
Current Directory  : /home/juhee/linux_lab3
Logged-in User     : juhee
```

Q5. Number Classification

```
juhee@kali:~/linux_lab3$ cat number_type.sh
#!/bin/bash
read -p "Enter an integer: " num

if [ $num -gt 0 ]; then
    echo "$num is Positive."
elif [ $num -lt 0 ]; then
    echo "$num is Negative."
else
    echo "The number is Zero."
fi

juhee@kali:~/linux_lab3$ chmod +x number_type.sh
juhee@kali:~/linux_lab3$ ./number_type.sh
Enter an integer: 15
15 is Positive.
juhee@kali:~/linux_lab3$ ./number_type.sh
Enter an integer: -7
-7 is Negative.
```

Q6. Even or Odd

```
juhee@kali:~/linux_lab3$ cat even_odd.sh
#!/bin/bash
if [ -z "$1" ]; then
    echo "Usage: $0 "
    exit 1
fi

if [ $(( $1 % 2 )) -eq 0 ]; then
    echo "$1 is Even."
else
    echo "$1 is Odd."
fi

juhee@kali:~/linux_lab3$ chmod +x even_odd.sh
juhee@kali:~/linux_lab3$ ./even_odd.sh 8
8 is Even.
juhee@kali:~/linux_lab3$ ./even_odd.sh
Usage: ./even_odd.sh
juhee@kali:~/linux_lab3$ echo $?
1
```

Q7. File and Directory Tests

```
juhee@kali:~/linux_lab3$ cat file_check.sh
#!/bin/bash
path="$1"
if [ -z "$path" ]; then
    echo "Usage: $0 "
    exit 1
fi

if [ -f "$path" ]; then
    echo "'$path' is a regular file."
    if [ -x "$path" ]; then
        echo "File is executable."
    else
        echo "File is NOT executable."
    fi
elif [ -d "$path" ]; then
    echo "'$path' is a directory."
else
    echo "'$path' is neither a regular file nor a directory (or does not exist)."
fi

juhee@kali:~/linux_lab3$ chmod +x file_check.sh
juhee@kali:~/linux_lab3$ ./file_check.sh data/names.txt
'data/names.txt' is a regular file.
File is NOT executable.
juhee@kali:~/linux_lab3$ ./file_check.sh data
'data' is a directory.
```

Q8. For Loop Practice

```
juhee@kali:~/linux_lab3$ cat multiplication.sh
#!/bin/bash
num=$1
if [ -z "$num" ]; then
    echo "Usage: $0 "
    exit 1
fi

echo "Multiplication Table for $num:"
for i in {1..10}; do
    echo "$num x $i = $((num * i))"
done

juhee@kali:~/linux_lab3$ chmod +x multiplication.sh
juhee@kali:~/linux_lab3$ ./multiplication.sh 5
Multiplication Table for 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Q9. Loop Through Files

```
juhee@kali:~/linux_lab3$ cat list_txt.sh
#!/bin/bash
echo "Listing all text files in 'data' directory:"
for file in data/*.txt; do
    if [ -f "$file" ]; then
        echo "Found text file: $file"
    fi
done

juhee@kali:~/linux_lab3$ chmod +x list_txt.sh
juhee@kali:~/linux_lab3$ ./list_txt.sh
Listing all text files in 'data' directory:
Found text file: data/names.txt
Found text file: data/notes.txt
Found text file: data/students.txt
```

Q10. While Loop

```
juhee@kali:~/linux_lab3$ cat countdown.sh
#!/bin/bash
count=10
while [ $count -ge 1 ]; do
    echo "$count"
    count=$((count - 1))
done
echo "Done!"

juhee@kali:~/linux_lab3$ chmod +x countdown.sh
juhee@kali:~/linux_lab3$ ./countdown.sh
10
9
8
7
6
5
4
3
2
1
Done!
```

Q11. Script Arguments

```
juhee@kali:~/linux_lab3$ cat arguments.sh
#!/bin/bash
echo "Script Name (\$0): $0"
echo "First Argument (\$1): $1"
echo "Second Argument (\$2): $2"
echo "Total Arguments Count (\$#): $#"
echo "All Arguments (\$@): $@"

juhee@kali:~/linux_lab3$ chmod +x arguments.sh
juhee@kali:~/linux_lab3$ ./arguments.sh Apple Banana
Script Name ($0): ./arguments.sh
First Argument ($1): Apple
Second Argument ($2): Banana
Total Arguments Count ($#): 2
All Arguments ($@): Apple Banana

juhee@kali:~/linux_lab3$ ./arguments.sh Red Green Blue
Script Name ($0): ./arguments.sh
First Argument ($1): Red
Second Argument ($2): Green
Total Arguments Count ($#): 3
All Arguments ($@): Red Green Blue
```

Q12. Exit Codes

```
juhee@kali:~/linux_lab3$ cat status.sh
#!/bin/bash
dir="$1"
if [ -d "$dir" ]; then
    echo "Directory '$dir' exists."
    exit 0
else
    echo "Directory '$dir' does not exist."
    exit 1
fi

juhee@kali:~/linux_lab3$ chmod +x status.sh
juhee@kali:~/linux_lab3$ ./status.sh data
Directory 'data' exists.
juhee@kali:~/linux_lab3$ echo $?
0
juhee@kali:~/linux_lab3$ ./status.sh non_existing_dir
Directory 'non_existing_dir' does not exist.
juhee@kali:~/linux_lab3$ echo $?
1
```

Q13. Output Redirection

```
juhee@kali:~/linux_lab3$ ls data > output/files.txt
juhee@kali:~/linux_lab3$ cat output/files.txt
names.txt
notes.txt
students.txt
juhee@kali:~/linux_lab3$ echo "--- Appended Section ---" >> output/files.txt
juhee@kali:~/linux_lab3$ ls data >> output/files.txt
juhee@kali:~/linux_lab3$ cat output/files.txt
names.txt
notes.txt
students.txt
--- Appended Section ---
names.txt
notes.txt
students.txt
```

Q14. Append a Log

```
juhee@kali:~/linux_lab3$ cat log_entry.sh
#!/bin/bash
read -p "Enter log message: " msg
timestamp=$(date "+%Y-%m-%d %H:%M:%S")
echo "[$timestamp] $msg" >> logs/activity.log
echo "Log entry appended."

juhee@kali:~/linux_lab3$ chmod +x log_entry.sh
juhee@kali:~/linux_lab3$ ./log_entry.sh
Enter log message: User logged in to system.
Log entry appended.
juhee@kali:~/linux_lab3$ ./log_entry.sh
Enter log message: Execution of Lab 3 commands complete.
Log entry appended.
juhee@kali:~/linux_lab3$ cat logs/activity.log
[2026-11-02 11:47:15] User logged in to system.
[2026-11-02 11:47:28] Execution of Lab 3 commands complete.
```

Q15. Input Redirection and Sorting

```
juhee@kali:~/linux_lab3$ sort < data/names.txt
Amit
Amit
Neha
Ravi
Zara
juhee@kali:~/linux_lab3$ sort -u < data/names.txt
Amit
Neha
Ravi
Zara
```

Q16. Error Logging

```
juhee@kali:~/linux_lab3$ ls non_existent_file.txt 2> logs/errors.log
juhee@kali:~/linux_lab3$ cat logs/errors.log
ls: cannot access 'non_existent_file.txt': No such file or directory
```

Q17. Combined Output and Error Logging

```
juhee@kali:~/linux_lab3$ ls data invalid_folder &> logs/combined.log
juhee@kali:~/linux_lab3$ cat logs/combined.log
ls: cannot access 'invalid_folder': No such file or directory
data:
names.txt
notes.txt
students.txt
```

Q18. Pipeline Practice

```
juhee@kali:~/linux_lab3$ ps aux | grep bash
juhee       1234  0.0  0.2  10544  4210 pts/0    Ss   11:30   0:00 bash
juhee       5678  0.0  0.1   9120  2450 pts/0    S+   11:47   0:00 grep bash
juhee@kali:~/linux_lab3$ ps aux | grep bash | wc -l
2
```

Q19. Backup Checker Script

```
juhee@kali:~/linux_lab3$ cat backup_check.sh
#!/bin/bash
dir="$1"
if [ ! -d "$dir" ]; then
    echo "Directory '$dir' does not exist."
    exit 1
fi

count=$(ls "$dir"/*.bak 2>/dev/null | wc -l)

if [ $count -gt 0 ]; then
    echo "Found $count backup file(s) in '$dir':"
    ls "$dir"/*.bak
else
    echo "No backup files found in '$dir'."
    echo "No backup files found in '$dir' on $(date)" >> logs/backup_log.txt
fi

juhee@kali:~/linux_lab3$ chmod +x backup_check.sh
juhee@kali:~/linux_lab3$ ./backup_check.sh backups
Found 2 backup file(s) in 'backups':
backups/db_01.bak  backups/db_02.bak
juhee@kali:~/linux_lab3$ ./backup_check.sh output
No backup files found in 'output'.
juhee@kali:~/linux_lab3$ cat logs/backup_log.txt
No backup files found in 'output' on Mon Nov  2 11:47:40 IST 2026
```

Q20. Integrated File Report

```
juhee@kali:~/linux_lab3$ cat file_report.sh
#!/bin/bash
if [ "$#" -ne 1 ]; then
    echo "Error: Exactly one filename argument is required."
    exit 1
fi

file="$1"

if [ -f "$file" ] && [ -r "$file" ]; then
    echo "=== File Analysis Report: $file ==="
    wc "$file"
    timestamp=$(date "+%Y-%m-%d %H:%M:%S")
    echo "[$timestamp] Processed file: $file successfully." >> logs/execution_history.log
else
    echo "Error: File '$file' does not exist or is not readable."
    timestamp=$(date "+%Y-%m-%d %H:%M:%S")
    echo "[$timestamp] Failed attempt on file: $file" >> logs/execution_history.log
    exit 1
fi

juhee@kali:~/linux_lab3$ chmod +x file_report.sh
juhee@kali:~/linux_lab3$ ./file_report.sh data/students.txt
=== File Analysis Report: data/students.txt ===
 4  4 52 data/students.txt
juhee@kali:~/linux_lab3$ ./file_report.sh non_existent.txt
Error: File 'non_existent.txt' does not exist or is not readable.
juhee@kali:~/linux_lab3$ cat logs/execution_history.log
[2026-11-02 11:47:50] Processed file: data/students.txt successfully.
[2026-11-02 11:47:55] Failed attempt on file: non_existent.txt
```